

OBSTETRICS

The effect of introduction of umbilical Doppler recordings to obstetric practice

*FRANK D. JOHNSTONE *Senior Lecturer*, **ROBIN PRESCOTT *Director of Medical Statistics*,
†PETER HOSKINS *Physicist*, ††IAN A. GREER *Professor*, ‡TOM MCGLEW *Senior Lecturer*,
§MARY COMPTON *Midwife*

*Department of *Obstetrics and Gynaecology, Centre for Reproductive Biology, and the Departments of **Public Health Science, †Physics, ‡Sociology, University of Edinburgh; ††Department of Obstetrics and Gynaecology, Royal Infirmary, Glasgow; §The Simpson Memorial Maternity Pavilion, Royal Infirmary of Edinburgh*

ABSTRACT

Objective To assess the effect on obstetric practice of clinician access to umbilical artery Doppler ultrasound results.

Design Randomised controlled trial.

Setting A large teaching hospital.

Subjects Two thousand two hundred and eighty-nine pregnancies defined as being at risk by referral for Doppler or fetal monitoring.

Interventions Continuous wave Doppler studies of umbilical artery. Results immediately available to clinicians.

Main outcome measures Fetal outcome: perinatal mortality, Apgar score and admission to the neonatal unit. Obstetric intervention: admission to hospital, induction of labour and caesarean section. Use of tests of fetal well being: cardiotocography, biophysical profile and ultrasound biometry.

Results The treatment and control groups were comparable in age, parity, gestation at point of entry and risk features. There were no overall differences in perinatal outcome, obstetric intervention or use of fetal monitoring. Examination of a subset recruited only because of hypertension or suspected intrauterine growth retardation ($n = 754$) similarly showed no difference attributable to group randomisation. Comparison of only those pregnancies retrospectively defined as low risk and high risk showed more use of cardiotocography in the high risk group with access to Doppler ($P = 0.007$) but no difference in the low risk group.

Conclusion Doppler umbilical artery recording has been shown to perform well in prediction power of antenatal fetal compromise. What has been examined in this study is the response of clinicians to the test. The results suggest that obstetricians do not use the test to modify their risk assessment, and, therefore, the need for fetal monitoring in particular pregnancies. There is a real need for accumulation of information from very large data sets, particularly in the prediction power of Doppler for antenatal fetal compromise from apparently chronic utero-placental cause to guide use of monitoring resources. If simply added to existing fetal monitoring techniques in a hospital where these are widely used, then umbilical artery Doppler recordings may at present simply involve extra resources of staff and expenses, without benefit.

Although Doppler recording of umbilical artery frequencies has aroused great research and clinical interest, its place in routine clinical management remains uncertain.

There are already effective biophysical tests of fetal wellbeing. In high risk pregnancy monitored by serial cardiotocography or biophysical profile, for example, fetal death is rare (Baskett *et al.* 1987; Oats *et al.* 1987; Divon *et*

al. 1990; Manning 1990), but these tests have disadvantages. They offer only short term prediction, have to be repeated frequently, and because ability to select truly at risk pregnancies is poor, large numbers of pregnancies may be monitored unnecessarily. This has obvious problems of patient inconvenience, satisfaction and expense as well as considerable total health system costs.

Doppler recording offers a different and complementary type of test. It may give long term warning of potential fetal compromise due to utero-placental cause, and is

Correspondence: Dr. F. D. Johnstone, Department of Obstetrics and Gynaecology, Centre for Reproductive Biology, 37 Chalmers Street, Edinburgh EH3 9EW, UK.

thus of potential value as a discriminator. Fetal compromise is common where there is absence of end diastolic frequency (AEDF) in the umbilical artery (Johnstone *et al.* 1988; McParland *et al.* 1990) but is rare, at least in non-diabetic pregnancy, when umbilical recordings are normal (Trudinger *et al.* 1991; Johnstone *et al.* unpublished data). This test could therefore improve the logical basis of obstetric management, segregating those pregnancies truly at risk (where delivery or intensive fetal monitoring is required) from those pregnancies not at high risk (where careful clinical care alone is adequate). This philosophy is implicit in many reports (Haddad *et al.* 1988a, b; Burke *et al.* 1990; Hofmeyr *et al.* 1991).

Although the above studies, together with other comparative studies (Nicolaidis *et al.* 1988; Tyrrel *et al.* 1989) are important and necessary preliminaries, they are an insufficient and essentially unrealistic way of assessing the clinical value of a new test. What is relevant is not simply the predictive power of the test, but the information given over and above that provided by routine clinical care. For example, in most cases of AEDF, even without this information, it is clinically obvious that there is a problem requiring delivery or close fetal monitoring. Whether Doppler provides useful additional information which improves obstetric management can only be assessed by randomised controlled trial. The need for randomised controlled trial is particularly germane to this technique because of unease about use of advanced technology in obstetrics and because of current economic stringency. As Paintin (1989) has stated 'all new methods of clinical management should be investigated and proved to be effective before coming into general use'.

In this hospital, we started using Doppler on a research basis in 1985, with a number of studies on methodology and clinical prediction power. During this early phase Doppler recordings were not available for clinical management. In 1988, although most clinicians favoured open access to Doppler measurements, there was sufficient collective clinical equipoise (Freedman 1987) about the overall effect of introduction of Doppler to justify randomised controlled trial. The objective of the trial was to assess the effect that clinician access to umbilical artery Doppler had on obstetric practice, without artificial restrictions on patient group or clinical management. The population studied was defined as women with pregnancies identified clinically as being at high risk. Our hypothesis was that those patients with access to Doppler, as compared with those who did not have access, would have at least as satisfactory a perinatal outcome, would have less fetal monitoring, would have more appropriate obstetric intervention, would be less likely to be admitted to hospital, and would involve less expense in their management. The particular end points specified were fetal mortality and morbidity, obstetric intervention and use of other tests of fetal monitoring (which are reported here) together with impact on obstetric decision making, health and personal costs, and patient satisfaction (which will be presented in a separate report).

Subjects and methods

The source of subjects for this trial was the total popula-

tion of women attending from 23 January 1989 to 31 August 1990. The criterion for entry into the trial was concern of medical staff about antenatal fetal wellbeing. Randomisation occurred for every woman where either there was a wish to refer for Doppler studies, or there was referral for antenatal fetal monitoring (cardiotocography or biophysical profile).

All women meeting these criteria were randomised regardless of risk group, but additional separate analysis was planned for the subgroup with hypertensive complications and suspected intrauterine growth retardation. A further separate analysis was planned for those pregnancies retrospectively classified as low risk or high risk.

Extensive discussion took place before the trial was planned. Doppler recordings initially had not been available for clinical management, but although still a research technique, recordings had been made available for nine months before the study started. Obstetricians were thus familiar with the use of test results. Several meetings occurred for medical and midwifery staff where information about the study, about results of previous studies and about use of Doppler was disseminated. A comprehensive collection of articles on umbilical artery Doppler was kept accessible in the day care unit.

The attitude of obstetricians before the trial was not formally assessed. In discussions, most considered a trial to be correct, and none considered the use of Doppler to be unethical. Only two obstetricians considered that Doppler should be available to everyone without further investigation. Overall, however, there was agreement, and a sufficient degree of collective clinical equipoise to justify a randomised controlled trial RCT.

The model used for the randomised controlled trial was that of Zelen (1979) where the addition of a new test is compared with the best contemporary techniques. After randomisation, half the women were offered Doppler, and the study was explained. The other women were not allowed access to Doppler under any circumstances. They were not specifically informed, but patient information sheets in the day care unit explained that an RCT of Doppler was occurring. The reason for this design was that allowing access to Doppler in some of the control group would fatally weaken design which would itself have ethical implications. The justification was the collective equipoise of the staff, the absence of information that Doppler aided clinical management, the fact that most obstetric units did not have Doppler, and that many units with Doppler still did not release results for clinical management. Ethical approval for the study was given by the Reproductive Medicine Ethical Committee.

Doppler ultrasound flow velocity waveforms were acquired using a continuous wave Doppler with a 4 MHz transducer and spectrum analyser, filter level 60–80 Hz (Doptek, UK). For umbilical artery recordings, the operator adjusted the probe position until the characteristic waveforms were seen with a clearly defined outline and free from venous or other interference. Waveforms were only accepted if a clear continuous signal corresponding to the umbilical vein was visible in the reverse channel, and a sequence of at least 15 cardiac cycles was usually viewed.

Usually routinely, and in all cases of difficulty, the operator located a loop of the umbilical cord with the assistance of a real time scanner. The extreme ends of the cord were avoided for sampling. All signals were recorded on to an audio cassette.

The resistance index (RI) (maximum systolic – end diastolic/maximum systolic) was manually calculated using the Doppler instrument's light pen system. This index was used because it is simple, and other mathematical descriptions of the waveform have not improved prediction power (Hoskins *et al.* 1989a). Normalisation of the index for heart rate within normal limits has not been found to be helpful (Hoskins *et al.* 1989b) and no adjustment for heart rate was therefore carried out. In most cases waveforms were acquired during fetal apnoea. These showed low variability and the average RI of five waveforms was taken. When it was impractical to wait for a period of fetal apnoea, waveforms were accepted during light fetal breathing. These demonstrated a small degree of variability and the average RI of 15 waveforms was taken. Highly irregular traces associated with strong fetal breathing were excluded (Hoskins *et al.* 1988).

Most recordings were made by the same person (MC). Initial studies showed a high degree of reproducibility, and regular assessment of waveform quality from hard copy was carried out (PH). Results of UARI were displayed against a normal range (mean $\pm$ 2SD) and this form was kept in the patient's file.

Sequentially numbered opaque sealed envelopes were attached by stapling to the case notes of all women attending this hospital. Randomisation was carried out by opening the envelope for every woman who met the criteria described above. A record was kept with addressograph labels of each woman and the envelope number was written on this. A master file was kept with each envelope number and Doppler allocation (medical statistics unit). Thus, even if an envelope was lost, the Doppler group for each woman could be traced. It also meant that deliberate attempts to cheat the system, by changing over envelopes from different patients, could be detected. Data analysis was on an intention-to-treat basis.

Management of pregnancy was determined by the consultant in charge of the patient and his team. Where UARI was greater than 4SD above the mean, then this information was given personally to the consultant in charge by MC. Absent end diastolic flow was considered to indicate either delivery or admission to hospital with frequent fetal monitoring, according to clinical conditions.

A database was written by Dr Stewart Irvine and Mrs Margaret Reid for use in the day care unit. Data on reason for testing, tests performed and results were entered for all women at the time of testing. This information was also printed out and filed in the notes. Information on pregnancy outcome was routinely recorded on the hospital computer. Data on number of ultrasound scans performed were obtained from the notes. Information about admission to the neonatal unit was obtained from admission records, and computer data. Centile birth weights were calculated after standardising for fetal sex,

maternal parity and gestation at delivery (Forbes *et al.* 1982). All data were downloaded and entered into the mainframe computer of the university.

All data processing and statistical analysis was performed using the SPSS-X package. Two sample *t*-tests and χ^2 tests were used to compare results from the two treatment groups. Based on anticipated numbers of pregnancies recruited, and interventions (from 1986–1987) the study had adequate power to test the hypothesis. For example, on expected numbers (an estimated 9200 pregnancies leading to the randomisation of 2300 high risk women) there was an 80% probability of showing statistical significance if there was an 8% decline in cardiotocography or an 11% decline in use of induction of labour.

Results

Two thousand two hundred and eighty-nine women were included in the study. A total of 8018 women were delivered in hospital during the same time. This suggests that approximately 28.5% of the total population was assessed as being at high enough risk to have fetal monitoring and be included in the trial.

At least one Doppler result above our normal range was recorded in 8% of women (7.5% of all 1823 recordings). However, recordings more than 4 standard deviations (SD) above the mean were found in only 0.9% of women (1.0% of all recordings). Abnormal results were recorded in 3.5% of the retrospectively defined low risk group but none were greater than 4SD above the mean. In the high risk group, 24.3% of women had abnormal Dop-

Table 1. Comparisons of age, parity, gestation at point of entry and risk features by allocated group.

Characteristics	Doppler <i>n</i> = 1114	Non Doppler <i>n</i> = 1175	Total <i>n</i> = 2289
Age (years), mean (SD)	26.6 (5.5)	26.6 (5.3)	26.6 (5.4)
Parity <i>n</i> (%)			
0	443 (40)	497 (42)	940 (41)
1	351 (32)	323 (28)	674 (30)
2	173 (16)	205 (18)	378 (17)
>2	147 (13)	149 (13)	296 (13)
Gestation at point of entry (weeks), mean (SD)	36.3 (4.9)	36.6 (4.5)	36.5 (4.7)
Main risk feature (<i>n</i>)			
Post dates	385	408	793
(with other risks)	(87)	(93)	(180)
Hypertension	251	258	509
(with other risks)	(80)	(66)	(146)
Reduced fetal movement	157	181	338
(with other risk)	(63)	(71)	(134)
Intrauterine growth retardation	135	136	271
(with other risks)	(44)	(47)	(91)
Bad previous obstetric history	45	36	81
Multiple pregnancy	18	22	40
Insulin dependent diabetes	20	21	41
Other	206	222	428

Table 2. Perinatal and related infant deaths by study group.

SB = stillbirth; PRM = premature rupture of membranes; NND = neonatal death.

A. Doppler available

	Gestation (weeks)	Birthweight (g)	Classification	Comments	Doppler results
Congenital abnormality					
1	34	1690	NND	Chromosome anomaly thalamic bleed	Normal
2	37	1130	SB	Congenital adrenal hypoplasia extreme intrauterine growth retardation	Abnormal (diastolic flow present)
3	29	750	NND	Multiple abnormalities	Not done
4	39	2440	NND	Truncus arteriosus (type 1) with absence of arch of aorta	Normal
Spontaneous preterm labour					
5	25	950	NND	PRM for 9 weeks pulmonary hypoplasia spontaneous labour	Normal
6	26	730	NND	PRM, chorioamnionitis, spontaneous labour	Not done
Twin-to-twin transfusion					
7	30	700	NND	Pulmonary hypoplasia	Abnormal (diastolic flow present)
8	30	1295	NND	Cardiac and renal failure	Abnormal (diastolic flow present)
Non-immunological hydrops					
9	29	2960	SB	Massive unexplained hydrops	Not done
Pre-eclampsia					
10	30	1258	NND	Fulminating pre-eclampsia. Delivered in maternal interest; Baby died ventricular arrhythmia	Normal
Placental abruption					
11	40	3265	SB	Large placental abruption. Normal pregnancy	Normal
Already dead					
12	30	520	SB	Already dead when tested	—

B. Doppler not available

Congenital abnormality					
1	37	2420	NND	Microcephaly, VSD abnormal gonads, excessive haemopoiesis, No syndrome identified	—
2	34	3340	NND	Severe cyanotic heart disease	—
3	35	1760	Infant death	Genetic disorders involving underdevelopment of the brain	—
4	39	2625	Infant death	Severe congenital ichthyosis (harlequin type)	—
5	33	2150	NND	Dystrophia myotonica	—
6	38	2270	NND	Trisomy 13	—
Spontaneous preterm labour					
7	27	710	NND	Chorioamnionitis spontaneous labour (twin)	—
8	29	1250	Infant death	Died after going home (twin)	—
9	26	915	Infant death	Prolonged PRM chorioamnionitis spontaneous labour	—
Pre-eclampsia					
10	28	856	NND	Severe hypertension, abnormal cardiotocography led to caesarean section	—
11	29	910	Infant death	Severe pre-eclampsia broncho pulmonary dysplasia	—
12	32	1510	SB	Severe pre-eclampsia placental abruption	—
Placental abruption					
13	27	770	NND	Placental abruption caesarean section	—
Already dead					
14	28	126	SB	Dead at time of assessment (twin)	—
15	29	1170	SB	Dead at time of assessment (twin of case 8)	—
Intrauterine death, IUGR					
16	35	1700	SB	Twin: growth retarded	—

pler results with 4.6% of them having results greater than 4SD above the mean.

Assessment was given as stipulated in the randomisation except for the following instances. Doppler measurement was not carried out despite the woman being randomised to the Doppler available group in 24 cases. This was because of urgent transfer to labour ward ($n = 5$); patient declined any fetal monitoring ($n = 2$); persistent fetal breathing ($n = 6$); fetal death ($n = 1$); cardiotocography carried out at night and patient transferred to labour ward by morning ($n = 2$); patient sent away by mistake ($n = 1$); and technical problems ($n = 3$). The reason was not clear for four women.

In three cases, Doppler measurement was performed despite the woman being randomised to the 'Doppler not available' group. In one pregnancy, the consultant in charge changed envelopes with that from another woman in order to enable the first woman to have Doppler. Doppler was carried out but because of the record system the switch was discovered immediately, this was publicised, and did not happen again. In one pregnancy (no Doppler) the consultant in charge insisted that Doppler should be carried out for research reasons. In a third pregnancy Doppler was carried out in error.

Table 1 shows that the treatment and control groups were comparable. The reasons for fetal assessment were past dates (35%), hypertensive complications (23%), reduced fetal movements (15%), intrauterine growth retardation (12%) and a number of other less frequent reasons. The proportion in each category with other risks specified is shown.

Table 2 shows that there was no significant difference in perinatally related mortality. There were only three still-

births not associated with fetal anomaly; two were due to placental abruption (one in each group). There was only one stillbirth associated with an apparent chronic uteroplacental process (growth retardation in a twin pregnancy). This case was in the non Doppler group.

Table 3 shows that there were no statistically significant differences in induction of labour, caesarean section, gestation at delivery, birthweight, Apgar score or admission to the neonatal unit. There were no differences in mean number of nights spent in hospital, either antenatally or postnatally.

Table 4 shows that 6996 cardiotocographs, 3813 biophysical profiles, and 2306 ultrasound scans to estimate fetal size were carried out. However, there were no statistically significant differences in use of these tests by study allocation group.

Table 5 shows results for only those pregnancies where the referral diagnosis was hypertension or intrauterine growth retardation. There were no substantial differences.

Table 6 shows only those pregnancies where birthweight was greater than 25th centile, delivery occurred vaginally, at term, with one minute Apgar greater than 7 and no admission to the neonatal unit (retrospectively classified as low risk), and those pregnancies where birthweight was less than 10th centile, or there was antenatal fetal compromise or caesarean section because of fetal distress in early labour (retrospectively classified as high risk). There are the expected differences between these groups. Use of Doppler does not appear to have altered the frequency of other fetal monitoring in the low risk group. However, use of Doppler is associated with increased fetal monitoring in the high risk group.

Table 3. Comparison of delivery details, admission to the neonatal unit and hospital admission time by study group. CS = caesarean section.

	Doppler <i>n</i> = 1114	Non Doppler <i>n</i> = 1175	Total <i>n</i> = 2289
Induction of labour, <i>n</i> (%)	334 (21)	371 (22)	715 (21)
Mode of delivery, <i>n</i> (%)			
Elective CS	63 (5.7)	73 (6.2)	136 (5.9)
Emergency CS	117 (10.5)	125 (10.6)	242 (10.6)
Total CS	170 (16.2)	198 (16.8)	378 (16.5)
Gestation at delivery (weeks), mean (SD)	39.5 (2.3)	39.4 (2.4)	39.4 (2.3)
Birthweight			
Singletons, <i>n</i>	1095	1147	2242
Mean (singletons)	3301 (656)	3289 (647)	3295 (651)
< 10th centile	122	122	244
< 25th centile	167	179	337
> 25th centile	799	849	1648
Total	1088	1141	2229
Apgar score			
$\bar{n}$	1128	1196	2324
Mean (SD)	8.8 (1.2)	8.8 (1.1)	8.8 (1.1)
< 7 at 5 min, <i>n</i> (%)	26 (2.3)	29 (2.4)	55 (2.4)
Admission to neonatal unit			
up to 48 h	10	7	17
> 48 h	86	94	180
Total, <i>n</i> (%)	96 (8.5)	101 (8.4)	197 (8.5)
Antenatal admission (nights), mean (SD)	3.6 (7.6)	3.7 (8.5)	3.6 (8.1)
Postnatal stay (nights), mean (SD)	4.2 (2.6)	4.4 (2.5)	4.3 (2.6)

Table 4. Comparisons of tests of antenatal fetal monitoring by study group.

Test	Doppler group <i>n</i> = 1114	Non Doppler group <i>n</i> = 1174	Total <i>n</i> = 2288
Doppler			
Number of tests			
0	24	1171	1195
1	827	3	830
2	122	—	122
3	49	—	49
>3	92	—	92
Mean (SD)	1.6 (1.8)	—	—
Antenatal cardiotocography			
Number of tests	3454	3542	6996
0	121	103	224
1	309	365	674
2	250	235	485
3	128	130	258
4	81	100	181
5	55	69	124
>5	163	163	326
Mean (SD)	3.1 (3.7)	3.0 (3.3)	3.1 (3.5)
Biophysical profile			
Number of tests	1926	1887	3813
0	391	449	840
1	343	359	702
2	155	151	306
3	70	62	132
4	44	43	87
5	28	25	53
>5	76	76	152
Mean (SD)	1.7 (2.8)	1.6 (2.7)	1.7 (2.7)
Ultrasound measurement of fetal size*			
Number of tests	1118	1188	2306
0	476	497	973
1	340	362	702
2	151	150	301
3	64	86	150
>3	60	55	115
Mean (SD)	1.0 (1.2)	1.0 (1.2)	1.0 (1.2)

*Abdominal and head circumference.

Discussion

This study was very strictly designed and conducted, and addressed the effect which introduction of umbilical artery Doppler would have on hospital practice. Study of a high risk obstetric population in which fetal monitoring was common did not show scope for improvement of perinatal mortality or morbidity. Our main hypothesis being tested, that the group with access to Doppler would have fewer tests of fetal monitoring, less intervention, and that the logical basis of clinical decision making would be improved, was not confirmed. Nor was there any adverse effect of use of Doppler, and overall no effect on total hospital management has been demonstrated.

Post hoc statistical power considerations show that initial estimates of the extent of morbidity and mortality were too high, largely because those were not as concen-

trated in our high risk group as anticipated. For example, only a 36% relative reduction in admission to the neonatal unit would have been detected with 80% power. Nevertheless, even this represents the ability to detect a reduction from a rate of 8.4% to 5.4%. Most endpoints suffered a similar loss, with, for example, the reduction in the number of cardiotocographs which would be detectable with 80% power being 14% in relative terms. In absolute terms, this is a reduction of 0.4 tests per woman. Although this is not as precise an answer as originally sought, it does exclude major differences in morbidity following the use of Doppler or substantial changes in the use of other investigations.

There are now three randomised controlled trials of use of Doppler which have been fully published in the obstetric literature (Trudinger *et al.* 1987; Tyrrel *et al.* 1990; Newnham *et al.* 1991). The first two studies are useful, but do not provide information which can readily be generalised to introduction of Doppler to an obstetric population. Trudinger *et al.* (1987) studied 300 hospital inpatients, but did show a statistically significant reduction in emergency caesarean section in spontaneous labour in the Doppler report group. However, their conclusion that use of Doppler improved obstetric decision making does not seem to be substantiated by the study data. Tyrrel *et al.* (1990) studied 500 women at high risk of intrauterine growth retardation or stillbirth. The intervention group had routine use of Doppler and fetal monitoring by biophysical profile. Important, statistically significant reductions in the frequency of depressed 5 minute Apgar scores and also the number of neonates with hypoxic ischaemic encephalopathy were found in the study group. However, this is impossible to interpret as far as use of Doppler is concerned because no information is given about use of fetal monitoring in the control group. The critical intervention could have been either Doppler or the widely accepted and established test of fetal monitoring. Newnham *et al.* (1991) studied 505 pregnant women who were referred to an ultrasound department for fetal investigation in the third trimester. Although there were small trends in obstetric management with allocation group, these were neither substantial nor statistically significant. The only statistically significant difference was that low Apgar scores were more common in the study group, the opposite finding of Tyrrel *et al.* (1990). Newnham *et al.* (1991) made the correct and important point that contribution of a new test to perinatal outcome would be affected by the availability of other biophysical tests of fetal wellbeing and the competence with which they are used in management.

Two further randomised controlled trials, not yet fully reported in the obstetric literature, have found use of Doppler to be associated with significant reductions in cases of intrauterine death (McParland & Pearce 1988; Omtzigt 1990).

Our study, the largest to date, complements data from these other studies, and directly addresses management aspects. Our philosophy has always been that Doppler assesses chronic placental circulatory reserve, whereas cardiotocography and biophysical profiles reflect current

Table 5. Comparison by study group including only those recruited because of suspected intrauterine growth retardation and/or maternal hypertension. CS = caesarean section.

Data	Doppler group	Non Doppler group	Total
Number of pregnancies	370	384	754
Age (years), mean (SD)	26.2 (5.5)	26.6 (5.3)	26.4 (5.4)
Gestation at point of entry (weeks), mean (SD)	34.4 (5.0)	34.9 (4.2)	34.7 (4.6)
Induction of labour, <i>n</i> (%)	156 (42)	161 (42)	317 (42)
Delivery method, <i>n</i> (%)			
Elective CS	24 (6.5)	33 (8.6)	57 (7.6)
Emergency CS	51 (13.8)	51 (13.3)	102 (13.5)
Total CS	75 (20.3)	84 (21.9)	159 (21.1)
Gestation at birth (weeks), mean (SD)	38.9 (2.2)	38.7 (2.6)	38.8 (2.4)
Birthweight (g), mean (SD)	3056 (719)	3018 (712)	3036 (715)
Mean Apgar <7, <i>n</i> (%)	9 (2.4)	13 (3.3)	22 (2.8)
Admission to neonatal unit, <i>n</i> (%)	46 (12.2)	46 (11.7)	92 (11.9)
Antenatal cardiotocography			
<i>n</i>	1554	1536	3090
Mean (SD)	4.2 (4.2)	3.9 (3.9)	4.1 (4.1)
Biophysical profile			
<i>n</i>	831	752	1583
Mean (SD)	2.3 (3.3)	2.0 (3.1)	2.1 (3.2)
Ultrasound measurement of fetal size (abdominal and head circumference)			
<i>n</i>	509	518	1027
Mean (SD)	1.4 (1.4)	1.4 (1.3)	1.4 (1.3)

fetal wellbeing. Doppler is complementary to these biophysical tests and its clinical value lies in its prediction power.

In pregnancies with some identified risk factor, the main role of umbilical artery Doppler will be to assist in planning management strategy: more aggressive (with delivery or intensive fetal monitoring) for the fetus with circulatory compromise and more conservative for the fetus with normal Doppler. In a population which is already having fetal monitoring, it seemed to us improbable that introduction of Doppler would improve perinatal outcome, although three trials have apparently demonstrated this. However, the hypothesis that management would be improved in terms of logical consistency, with an overall reduction in other fetal monitoring, seemed plausible. Not only was this hypothesis not sustained, but the trend was for more fetal monitoring to be performed in the group allocated to Doppler availability. The majority of women had only one Doppler study, but partly this was determined by the number of women recruited because of short term problems, such as post dates or reduced fetal movements. Even when consideration is restricted to women recruited because of suspected intrauterine growth retardation or hypertension, the results are similar. The study design did not select particular patient groups or specify particular protocols. This was partly because specifications of management based exclusively on test results would not be acceptable to clinicians (who have to take many aspects relating to the pregnancy and the woman into account) or to pregnant women (who enter into a decision based on all available information). It was also partly because choice of management could be subsequently revealed to be incorrect; and partly because results of the study could simply become self ful-

filling. Finally, this would not resolve the question of what effect introduction of Doppler would have on overall management.

The contribution of umbilical artery Doppler recordings to clinical practice was examined in different ways. Even though overall fetal monitoring and intervention was the same in both study groups, this could conceal very important benefits. Thus, because of the discriminatory power of Doppler, those pregnancies at low risk might have been subjected to less fetal monitoring and intervention in the Doppler group, whereas those pregnancies at high risk might have been more clearly recognised and had much more (and appropriate) fetal monitoring and intervention. In theory, these two effects could balance out in the total group with access to Doppler, so that there was no overall difference by study group. This model has been examined by defining, retrospectively, low risk and high risk groups. There was a statistically significant tendency to use more fetal monitoring in those in the high risk group with access to Doppler, but there was no corresponding decline in fetal monitoring in the low risk group.

One possible interpretation of the study result is as follows. The test continued to perform well in prediction (of antenatal fetal compromise), our total series now being in excess of 2000 high risk pregnancies (unpublished data). What has been examined in this study is the response of clinicians to the test. Apparently, despite the availability of a monitoring test with a burgeoning literature supporting its predictive value, obstetricians still do not use the test to modify their risk assessment, and therefore the need for further monitoring in particular pregnancies. The results of the study suggest that clinicians still rely on the clinical risk factors, which precipitated monitoring in the first place, to guide them in the subsequent need for,

Table 6. Retrospective review category.

Data	Low risk			High risk		
	Doppler <i>n</i> = 514	Non Doppler <i>n</i> = 532	TOTAL <i>n</i> = 1046	Doppler <i>n</i> = 152	Non Doppler <i>n</i> = 156	TOTAL <i>n</i> = 308
Age (years), mean (SD)	26.7 (5.6)	26.4 (5.1)	26.5 (5.4)	26.3 (5.7)	26.0 (5.7)	26.1 (5.7)
Nulliparous (%)	30	36	33	47	48	48
Gestation at point of entry (weeks), mean (SD)	36.1 (4.8)	36.6 (4.3)	36.4 (4.5)	34.2 (4.7)	34.7 (4.6)	34.5 (4.7)
Main risk features (%)						
Post dates	41.7	43.6	42.6	20.7	19.3	20.0
Hypertension	18.3	15.5	16.9	24.7	25.2	24.9
Reduced fetal movement	15.2	19.7	17.5	10.0	11.0	10.5
Intrauterine growth retardation	5.8	6.8	6.3	40.0	32.2	36.1
Other only	26.9	23.7	25.3	19.3	22.6	21.0
Onset of labour (%)						
Spontaneous	69.3	70.9	70.1	50.6	53.8	52.3
Induction	30.7	29.1	29.9	39.5	34.0	36.7
Elective CS	—	—	—	9.9	12.2	11.0
Mode of delivery (%)						
Spontaneous	100	100	100	43.4	51.9	47.7
Forceps/breech	—	—	—	15.2	5.8	10.3
Caesarean section	—	—	—	41.5	42.3	41.8
Gestation at birth (weeks), mean (SD)	40.1 (1.2)	40.0 (1.3)	40.0 (1.3)	38.6 (2.7)	38.6 (2.7)	38.6 (2.7)
Birthweight (g), mean (SD)	3588 (400)	3587 (412)	3587 (406)	2543 (689)	2571 (655)	2558 (671)
Mean Apgar score at 5 min (SD)	9.0 (0.2)	9.0 (0.1)	9.0 (0.2)	8.6 (1.6)	8.7 (1.2)	8.6 (1.4)
Apgar <7 at 5 min	0	0	0	6	5	11
Neonatal unit admission (%)	0	0	0	15.2	14.9	15.1
Cardiotocography, mean (SD)	2.6 (2.9)	2.6 (2.8)	2.6 (2.8)	4.5 (4.6)	3.2 (3.7)	3.9* (4.2)
Biophysical profile, mean (SD)	1.2 (1.7)	1.3 (2.2)	1.2 (1.9)	3.2 (4.4)	2.4 (3.1)	2.8 (3.8)
Ultrasound biometry, mean (SD)	0.74 (1.0)	0.74 (1.0)	0.74 (1.0)	1.7 (1.4)	1.6 (1.3)	1.6 (1.3)

**t* = 2.7; *P* = 0.007.

and frequency of, monitoring, rather than making use of the information from Doppler. This may be of concern as many of the risk factors which will have precipitated monitoring will be very weak or poorly defined. Attitudes to monitoring strategy have been studied, both in junior staff, who initiate more tests, and consultants; this data will be prepared for publication (McGlew *et al.* unpublished data).

It is possible that a new test is not readily accepted by clinicians and it may be that practice will only change after several years' experience. Such a change could be due to obstetricians becoming convinced of the role of the test or could simply be due to increasing familiarity. What this may imply is that we are at the start of a learning curve, and if the study were to be repeated in a few years' time, a different outcome would be found. However, it may also mean that currently available information on prediction power is not accepted as sufficient to override clinical concern. Published studies on umbilical artery Doppler often have small numbers, specialised conditions, and end points, such as birthweight, which are not of primary concern. There is a real need for further information from very large data sets, particularly on the incidence of antenatal fetal compromise (and especially fetal death) from apparently chronic utero-placental cause in pregnancies

where umbilical artery Doppler recordings were normal. Sufficient information is not currently available.

Our conclusion is that, if simply added to existing fetal monitoring techniques in a hospital in which these are widely used, then umbilical artery Doppler recordings may simply involve extra resources of staff and expense without benefit. If such a technique is to be introduced, then sufficient evidence of its value should be available to staff to agree on criteria to be adopted for its role in clinical management.

Acknowledgments

This study was supported by SHHD Biomedical Grant Support. We are grateful to all our obstetric and midwifery colleagues who helped conduct this study. We would also like to thank Dr S. Irvine (Obstetrician) for writing the programme for the database for the day care unit; Mrs M. Reid (Computer Programmer) for her help with this and for facilitating transfer of files from the Mentor computer to the university main frame; Mrs A. Rennie and Miss G. White for clerical help at several points in the study; Mrs W. Hepburn for her expert preparation of the data; and Sister F. Dagge Bell for her support and help in

the day care unit. We are also grateful to all the pregnant women who took part in the study.

References

- Baskett T. F. *et al.* (1987) Fetal biophysical profile and perinatal death. *Obstet Gynecol* **70**, 357–360.
- Burke G. *et al.* (1990) Is intrauterine growth retardation with normal umbilical artery blood flow a benign condition? *Br Med J* **300**, 1044–1045.
- Divon M. Y., Lieblich R. M. & Girz B. A. (1990) Semi-weekly testing with zero false negative rate. Society of Perinatal Obstetricians. Tenth annual meeting, Houston, Texas, January 1990. Abstract 333, p. 349.
- Forbes J. F., Smalls M. J. & Cole S. K. (1982) Centile values of birthweight for gestational age in Scottish infants 1975–1979. Social Paediatrics and Obstetrics Research Unit, University of Glasgow.
- Freedman B. (1987) Equipoise and the ethics of clinical research. *New Engl J Med* **317**, 141–145.
- Haddad N. G. *et al.* (1988a) Umbilical artery Doppler flow velocity waveform analysis and the outcome of hypertensive pregnancies. *J Obstet Gynecol* **9**, 9–13.
- Haddad N. G. *et al.* (1988b) Umbilical artery Doppler waveform in pregnancies with uncomplicated intrauterine growth retardation. *Gynaecol Obstet Invest* **26**, 206–210.
- Hofmeyr G. J. *et al.* (1991) Umbilical artery resistance index as a screening test for fetal well being II Randomised feasibility study. *Obstet Gynecol* **78**, 359–362.
- Hoskins P. R. *et al.* (1988) The determination of the presence of fetal apnoea using umbilical artery and umbilical vein Doppler waveforms. *J Ultrasound Med Biol* **14**, 589–592.
- Hoskins P. R. *et al.* (1989a) The choice of index for umbilical artery Doppler waveform. *J Ultrasound Med Biol* **15**, 107–111.
- Hoskins P. R. *et al.* (1989b) Heart rate variation of umbilical artery Doppler waveform. *J Ultrasound Med Biol* **15**, 101–105.
- Johnstone F. D. *et al.* (1988) Umbilical artery Doppler flow velocity waveform: the outcome of pregnancies with absent end diastolic flow. *Eur J Obstet Gynecol Reproduct Biol* **28**, 171–178.
- Manning F. A. (1990) Fetal biophysical profile score: current status. *Obstet Gynecol Clin N Amer* **17**, 147–162.
- McParland P. & Pearce J. M. (1988) Review: Doppler blood flow in pregnancy. *Placenta* 427–450.
- McParland P., Steel S. & Pearce J. M. (1990) The clinical implications of absent or reversed end-diastolic frequencies in umbilical artery flow velocity waveforms. *Eur J Obstet Gynecol Reproduct Biol* **37**, 15–23.
- Newnham J. P. *et al.* (1991) Doppler flow velocity analysis in high risk pregnancies: a randomised controlled trial. *Br J Obstet Gynaecol* **98**, 956–963.
- Nicolaidis K. H. *et al.* (1988) Absence of end diastolic frequencies in umbilical artery: a sign of fetal hypoxia and acidosis. *Br Med J* **297**, 1026–1027.
- Oats J. N., Chew F. T. K. & Ratten V. J. (1987) Ante-partum cardiotocography—an audit. *Aust NZ J Obstet Gynaecol* **27**, 82–86.
- Omtzigt A. W. J. (1990) Clinical value of umbilical Doppler velocimetry. Thesis, Utrecht, The Netherlands.
- Paintin D. B. (1989) What this journal would like to publish. *Br J Obstet Gynaecol* **96**, 377–382.
- Trudinger B. J. *et al.* (1987) Umbilical artery flow velocity waveforms in high-risk pregnancy: randomised controlled trial. *Lancet* **i**, 188–190.
- Trudinger B. J. *et al.* (1991) Fetal umbilical artery velocity waveforms and subsequent neonatal outcome. *Br J Obstet Gynaecol* **98**, 378–384.
- Tyrrel S. N., Lilford R. J. & Obaid A. H. (1989) Relationship between umbilical artery Doppler waveforms and cord gases. *Obstet Gynecol* **74**, 332–337.
- Tyrrel S. N. *et al.* (1990) Randomised comparison of routine vs highly selective use of Doppler ultrasound and biophysical scoring to investigate high risk pregnancies. *Br J Obstet Gynaecol* **97**, 909–916.
- Zelen M. (1979) A new design for randomised clinical trials. *New Engl J Med* **300**, 1242–1245.

Received 18 May 1992

Accepted 15 December 1992